Nowcasting Real GDP Growth in The Caribbean

Preliminary results for Barbados

Jose Luis Saboin
Diego Guerrero
Angelo Mazzocca

Inter-American Development Bank & Northwestern University

June 2026

Objectives & Agenda

Objectives

- Show the tool
- Obtain feedback
- Discuss the tool's adoption by Central Bank, Ministry of Finance and Statistical Office partners

Agenda

- Presentation (30 min)
- Q & A (15 min)
- Next steps (15 min)

Motivation: The data timeliness problem...

- Barbados Q4 2025 GDP released on March 28, 2026 — **3 months** after the quarter ended
- Globally: Only 55% of countries publish quarterly GDP timely (Berry et al., 2018)
- Critical for small, open economies vulnerable to:
  - Natural disasters
  - Other external shocks
- Also important for timely decision-making
- **Solution**: Nowcasting with high-frequency indicators

Motivation: ...coupled with access challenges

TABLE 2: GROSS DOMESTIC PRODUCT BY ECONOMIC ACTIVITY (B$ MILLIONS)

REAL I Quarter 2015 - II Quarter 2024 (2018 = 100)

Industry Classification (Revision 4)	ISIC Code	Q1 2023 R	Q2 2023 R	Q3 2023 R	Q4 2023 R	Q1 2024 P	Q2 2024 P
Agriculture, forestry and fishing	A	16.0	13.1	15.3	23.1	23.3	19.8
Mining	B	24.9	26.1	27.4	23.2	32.5	28.1
Manufacturing	C	12.4	11.4	19.4	22.2	11.8	10.1
Electricity and gas, Water supply and sewerage	D & E	59.8	57.8	62.2	46.6	48.0	51.8
Construction	F	160.8	155.8	169.3	147.7	131.6	241.8
Wholesale and retail trade, Motor Vehicle repairs	G	331.2	325.3	304.1	305.6	306.9	309.7
Transport and Storage	H	152.9	161.0	198.3	180.8	162.1	180.5
Accomodation and Food Services	I	269.1	287.0	286.5	265.9	285.0	313.9
Information and Communication	J	115.4	100.4	121.2	120.3	128.2	129.6
Financial and Insurance activities	K	261.9	253.4	232.8	235.1	231.3	240.4
Real Estate activities	L	529.6	520.0	508.2	507.1	536.8	500.5
Professional, scientific and technical services	M	125.8	129.5	114.0	123.8	123.7	126.4
Administrative and support services	N	84.3	84.0	78.4	79.9	83.4	82.8
Public administration and defense, Social security	O	255.1	257.3	255.9	254.6	253.4	252.3
Education	P	79.1	79.6	79.5	79.5	79.4	79.3
Human health and social work	Q	96.8	99.1	99.7	99.8	99.5	99.2
Arts, Other Services, Household Employment, Extraterritorial Org.	R,S&T	289.0	321.3	291.9	299.5	323.6	308.7
Total GDP at Basic Prices		2,864.2	2,882.0	2,864.2	2,814.7	2,860.8	2,974.9
Taxes less subsidies on products		258.5	207.2	256.1	186.5	223.1	262.8
Sub Total		3,122.7	3,089.2	3,120.2	3,001.2	3,083.8	3,237.7
Statistical descrepancy		125.9	124.7	126.1	121.3	124.7	130.9
GDP by economic activity at purchaser current prices		3,248.6	3,213.9	3,246.3	3,122.5	3,208.5	3,368.7
Constant Growth Rate (PREVIOUS QTR)		3.6%	-1.1%	1.0%	-3.8%	2.8%	5.0%
COMPARED TO THE SAME QUARTER IN PREVIOUS YEAR		8.32%	1.38%	1.55%	-0.41%	-1.23%	4.81%

(a) BNSI's National Accounts Table

Table 8.18 Selected Economic Indicators

I N D I C A T O R	Unit	2017	2018	2019	2020	2021	2022	2023	2024
Nonoil Exports (f.o.b.)	B$000	400,627	411,821	457,867	276,399	358,851	424,831	687,250	663,487
Nonoil Imports (c.i.f.)	B$000	2,874,958	2,938,016	2,551,720	1,818,573	2,824,819	3,153,452	3,562,393	4,139,121
Average Retail Price Index	Nov. 2014=100	103.20	105.54	108.17	108.21	111.35	117.59	121.19	121.68
Total Tourist Arrivals	(000)	6,136	6,622	7,250	1,795	2,101	7,001	9,655	11,217
Value of Construction Permits*	B$000	423,408	664,626	705,873	686,715	706,469	988,762	915,565	541,498
Value of Construction Starts*	B$000	136,615	118,174	102,881	154,564	313,714	340,161	212,104	169,738
Value of Construction Completions*	B$000	1,493,607	333,784	212,978	200,051	307,293	435,453	391,530	245,054
Government Revenue (Calendar Year)[P]	B$000	2,087,215	2,173,343	2,503,238	1,666,737	2,369,160	2,730,278	2,899,257	3,208,007
Government Revenue (Fiscal Year: Jul-Jun)[P]	B$000	2,070,259	2,042,385	2,426,318	2,082,003	1,908,776	2,605,701	2,855,445	3,069,106
Government Expenditure (Calendar Year)[P]	B$000	2,709,709	2,510,709	2,754,918	3,032,653	3,250,939	3,448,513	3,414,734	3,547,384
Government Expenditure (Fiscal Year: Jul-Jun)[P]	B$000	2,730,986	2,457,286	2,645,584	2,920,514	3,243,583	3,327,394	3,390,028	3,263,116
Government Debt (Direct Charge)[P]	B$000	7,180,089	7,498,912	7,733,214	9,417,448	10,317,372	11,035,946	11,427,480	11,767,607
Average Treasury Bill Discount Rate	%	1.89	1.71	1.75	1.93	2.85	2.88	2.91	2.94
Money Supply (M1)	B$000	2,654,036	2,728,160	3,248,398	3,472,120	3,715,501	4,296,761	4,318,660	4,626,451
Money Supply (M2)	B$000	6,763,155	6,707,279	7,304,977	7,505,689	7,772,682	8,459,695	8,602,888	8,989,337
Money Supply (M3)	B$000	7,037,296	7,108,822	7,892,847	7,864,180	8,220,340	9,002,044	9,133,858	9,485,830
Bank Credit (all currencies)	B$000	8,838,328	8,911,192	8,957,100	8,614,408	8,928,992	9,079,562	9,376,312	9,871,098
Bank Deposits (all currencies)	B$000	6,925,956	6,913,198	7,727,494	7,731,526	7,976,839	8,716,755	8,911,633	9,181,146

SOURCE: Data compiled from various tables in the Digest.
NOTE: * Excludes Family Islands' Statistics.

(b) CBOB's Quarterly Statistical Digest Table

- Complex PDF formats
- Scattered publications
- Manual processing required

Why Nowcasting?

- **Nowcasting**: Particularly valuable when (Akbal et al., 2023):
  - Official data has long publication lags
  - Data comes in complex PDF formats and publications are scattered
  - High-frequency indicators are available
  - Elevated uncertainty
  - Vulnerability to external shocks
- **Our contribution**: First *comprehensive* nowcasting framework for a Caribbean economy

Contributions

- **Comprehensive Caribbean framework**: first nowcasting toolkit applied across The Bahamas, Barbados, Jamaica, and Barbados
- **Unconventional data**: integration of Google Trends and satellite nighttime-lights (NTL) alongside official statistics
- **14 estimators:** spanning classical econometrics, regularization, and machine learning
- **Automation**: automated mining of conventional and unconventional data, automated data selection algorithm, automated model training, testing, performance evaluation and outputs (e.g., charts, tables, etc.)
- **Distributional output**: instead of a single point estimate, we produce a *distribution of nowcasts* across variable groups, specifications, estimators, and information vintages
- **Real-time relevance**: 2–8 month lead on official GDP releases

Main Findings

- We produce a **distribution of GDP-growth nowcasts** from multiple variable groups, estimators (14), and vintages (5)
- **Performance variation** (RMSE): performance varies across variable groups, estimators, and vintages; some specifications and models are consistently better
- **"Mixed"** groupings combining structured and unstructured indicators perform strongest
- **Google Trends** and **NTL** data boost nowcasting performance and are very timely
- Point forecasts within $\sim$**75% of actual GDP growth** using representative estimates from the nowcast distribution

Case Study: The Bahamas — Data integration

Comprehensive Dataset Structure (359 variables total)

Structured Data (218 series)

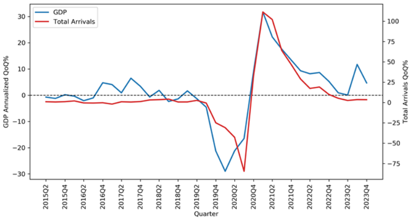
- Tourist arrivals
- Fiscal accounts
- Trade flows
- Monetary and financial indicators

Unstructured Data (141 series)

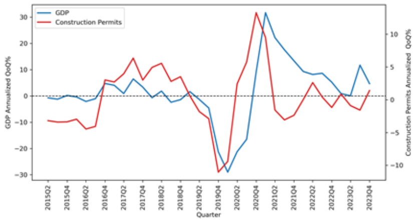
- Google Trends (101 variables)
- Satellite NTL (40 variables)

Case Study: The Bahamas — Structured data

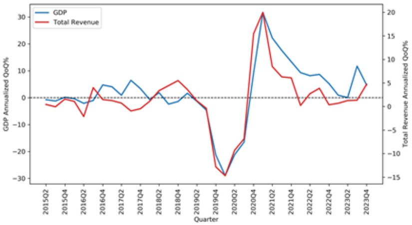
- Official statistics from BNSI (Bahamas National Statistical Institute) and CBOB (Central Bank of The Bahamas)
- Categories:
  - Tourist arrivals (air + sea)
  - Central government revenue and expenditure
  - Trade balance and foreign reserves
  - Monetary aggregates and credit
  - Electricity generation and selected sectoral activity
- Mixed monthly / quarterly frequencies aligned via vertical realignment



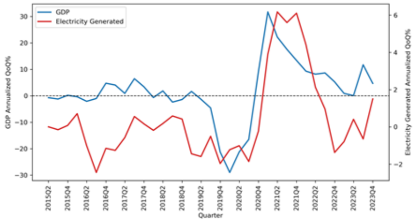
(a) Tourist Arrivals



(b) Construction Activity

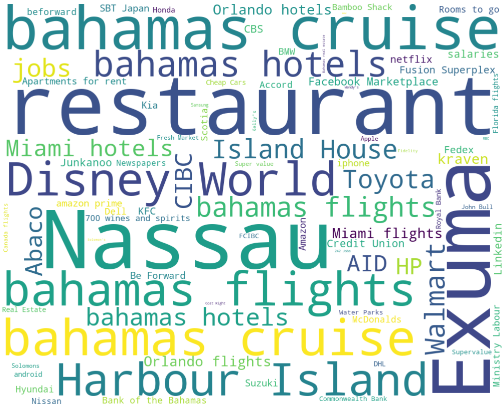


(c) Government Revenue



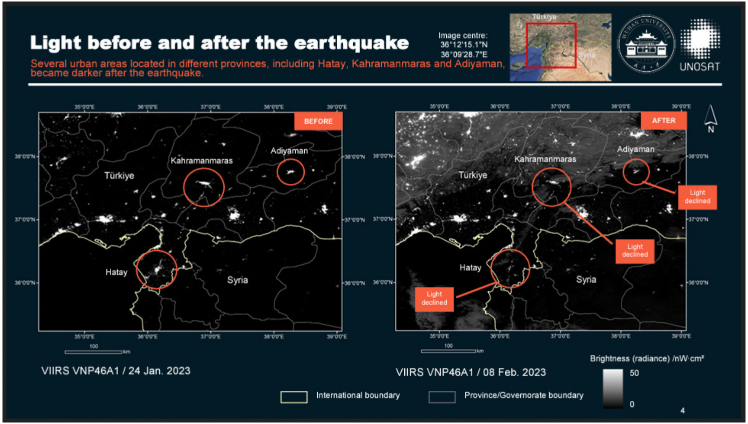
(d) Electricity Generation

- 101 search variables tracking consumer demand
- Categories: tourism, retail, jobs, services
- Real-time data with 1-week lag



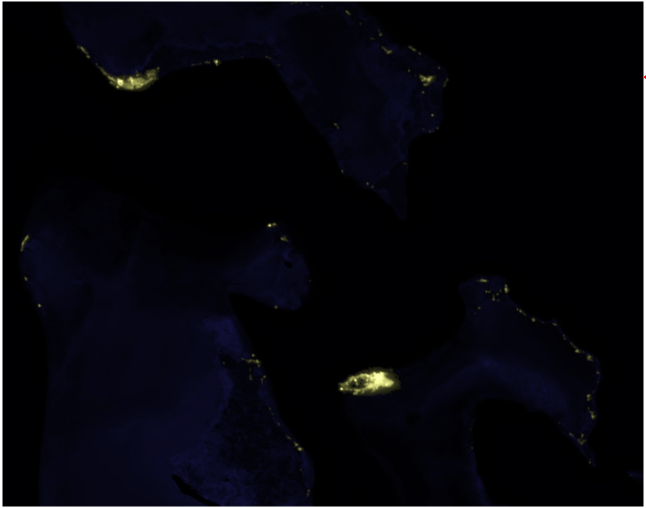
Word cloud: keywords sized by correlation with GDP

Case Study: The Bahamas — Unstructured data — Nighttime Lights



- Satellite-derived radiance is a real-time proxy for economic activity
- NASA / NOAA VIIRS daily product
- Source: UNDSAT

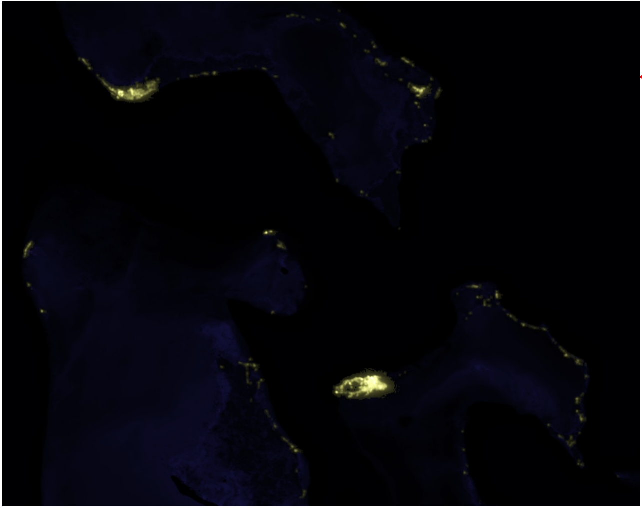
Case Study: The Bahamas — Unstructured data — Nighttime Lights



2019 baseline. Source: `blue-marble.de/nightlights`

The Island of Abaco right after Hurricane Dorian

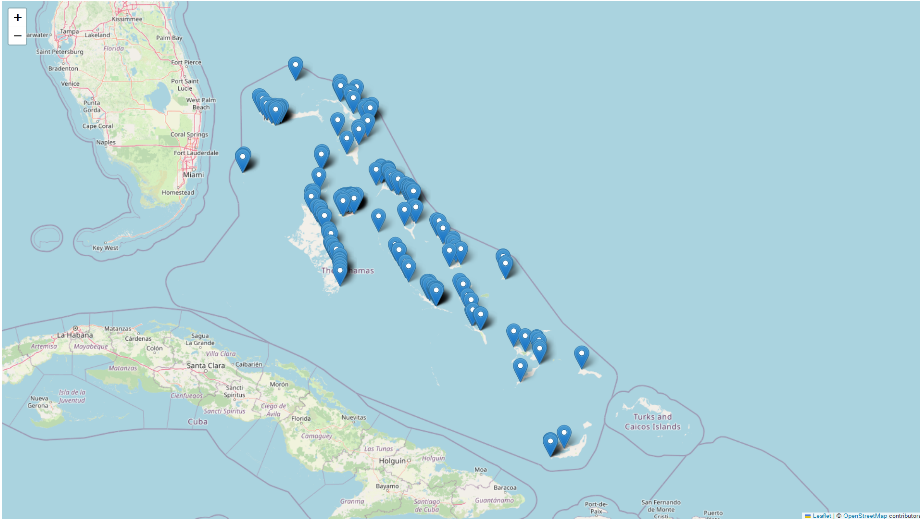
Case Study: The Bahamas — Unstructured data — Nighttime Lights



2023. Source: blue-marble.de/nightlights

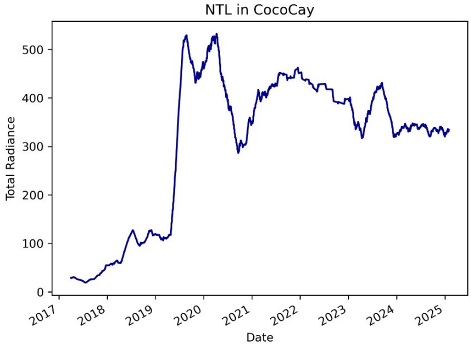
The Island of Abaco 4 years later

Case Study: The Bahamas — Unstructured data — Nighttime Lights



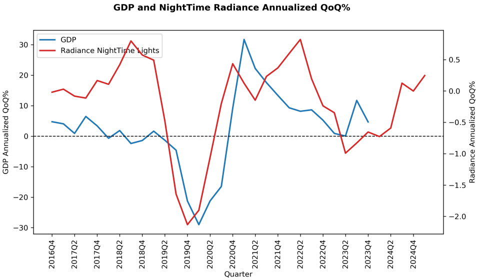
Case Study: The Bahamas — Unstructured data — Nighttime Lights

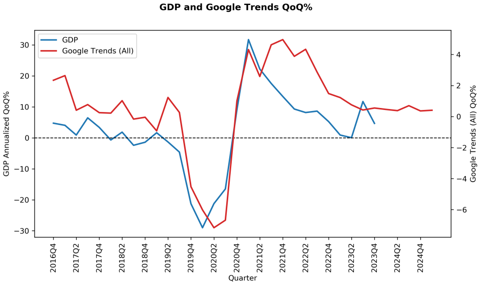
- **Radiance** (Watts/cm^2/sr) is the Sum of Lights (SOL) per pixel
- SOL adds all radiance across coordinates in a given day
- Ideal for comparing *over time*, not for cross-sectional magnitudes
- See World Bank, *Open Lights*



NTL total radiance in CocoCay (The Bahamas)

Case Study: The Bahamas — Unstructured data and GDP

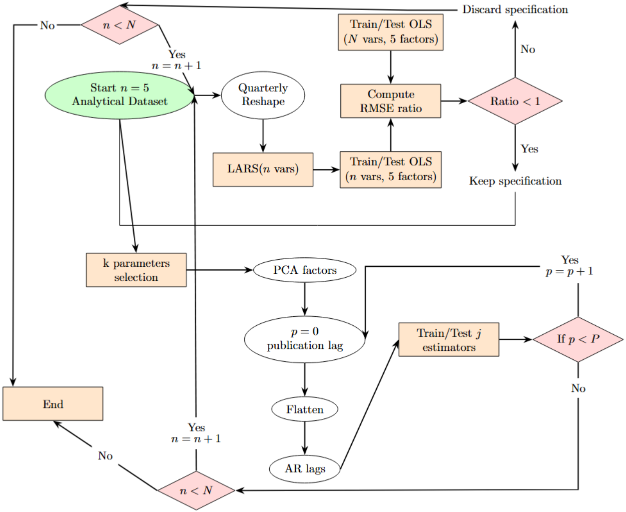




Selected unstructured indicators vs The Bahamas' real GDP growth

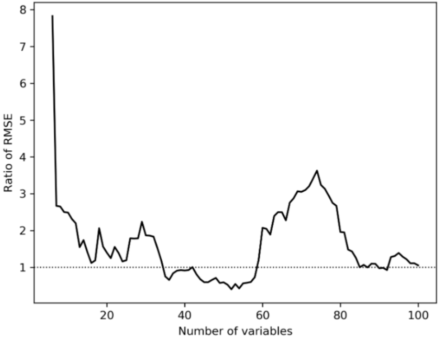
Methodology: Multi-Step Nowcasting Framework

- Variable pre-selection via LARS
- Multiple specifications (PCA factors, AR lags, flattening)
- 14 diverse estimators from econometrics and ML
- Distribution of nowcasts → point forecast + uncertainty



Variable pre-selection: Least Angle Regression (LARS)

- **Purpose**: handle high-dimensional data (359 variables)
- **Process**: LARS incrementally adds the most-correlated predictors in an equiangular direction
- Compare RMSE: 5-factor OLS with k vars vs. 5-factor OLS with all K vars; keep groupings that outperform the benchmark
- **Result**: **28 variable groupings** retained for The Bahamas
- Superior to other selection methods (Chinn et al., 2023)



LARS RMSE-ratio profile vs benchmark (The Bahamas)

Specifications: Key sources of variation (The Bahamas)

Specifications vary systematically:

1. **Flattening lags** to handle mixed frequencies:
   1.1 3 lags (baseline) — 28 specifications
   1.2 4 lags for linear / 5 lags for ML (expanded) — 28 specifications
2. **PCA factors** for dimensionality reduction:
   2.1 No factors — 56 specifications
   2.2 3 factors — 56 specifications
   2.3 5 factors — 56 specifications
3. **AR lags** to capture macroeconomic dynamics:
   3.1 No lags — 28 specifications
   3.2 2 lags — 28 specifications

Total: 197 specifications addressing core nowcasting challenges

Includes 1 human-expert specification for benchmarking

Comprehensive model collection: 13 estimators

Econometric Models

- ARMA
- OLS
- MIDAS (Ghysels et al., 2016)
- DFM (Bok et al., 2018)
- BVAR (Ankargren and Jonéus, 2020)

Regularization Models

- Ridge (Hoerl and Kennard, 1970)
- LASSO (Tibshirani, 1996)
- Elastic Net (Zou and Hastie, 2005)

Machine Learning Models

- Decision Trees (Breiman et al., 1984)
- Gradient Boosted Tree (Friedman, 2001)
- Random Forest (Breiman, 2001)
- LSTM (Hochreiter and Schmidhuber, 1997)
- MIDAS-ML (Babii et al., 2022)

We focus on the regularization and ML subset for this presentation.

Testing: Assessing accuracy through the nowcast cycle

- Five distinct time horizons (Chinn et al., 2023; Hopp, 2024)
- From 2 months before to 2 months after quarter end
- Accounts for asynchronous data publication
- Tracks nowcast improvement with information flow

Technical challenges addressed

Curse of Dimensionality

- LARS variable pre-selection (Bai and Ng, 2008; Chinn et al., 2023)
- PCA (Stock and Watson, 2002; Chinn et al., 2023)

Macroeconomic Dynamics

- AR lags of GDP (Jansen et al., 2016)

Mixed Frequencies & Ragged Edges

- Vertical realignment (Hopp, 2024)

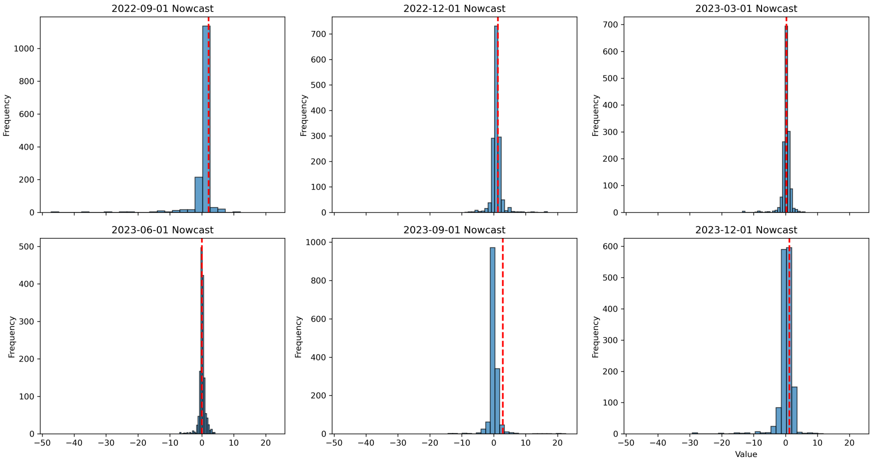
Model Uncertainty

- Econometrics + ML (Hopp, 2024; Kant et al., 2025)
- Model weighting (Kitchen and Monaco, 2003)
- Distribution of nowcasts

Key innovation: ensemble distribution approach (The Bahamas)

- **Traditional approach**: a single "best" model specification
- **Our approach**: a distribution of nowcasts from multiple models
- **Process**:
  1. Generate **1,858 estimates** (groupings × specifications × estimators)
  2. Filter: keep only models with RMSE below the median
  3. Result: distribution with confidence intervals
  4. Weight: inverse-RMSE weighting for final ensemble
- **Advantage**: captures uncertainty, leverages diverse model strengths

Case Study: The Bahamas — Results — Nowcast distribution



$N = 2{,}590$ estimates: 37 variable groups $\times$ 14 estimators $\times$ 5 vintages

Case Study: The Bahamas — Results — Inverted-RMSE weighted nowcast

Inverted-RMSE weighted nowcast
of all quarterly estimates by vintage

Quarter	GDP	Zero-back	Two-ahead
2022 Q2	1.99	1.44	1.67
2022 Q3	2.09	1.19	1.44
2022 Q4	1.29	0.95	1.05
2023 Q1	0.23	0.35	0.22
2023 Q2	0.01	0.20	0.13
2023 Q3	2.81	0.25	0.34
2023 Q4	1.15	0.64	0.37

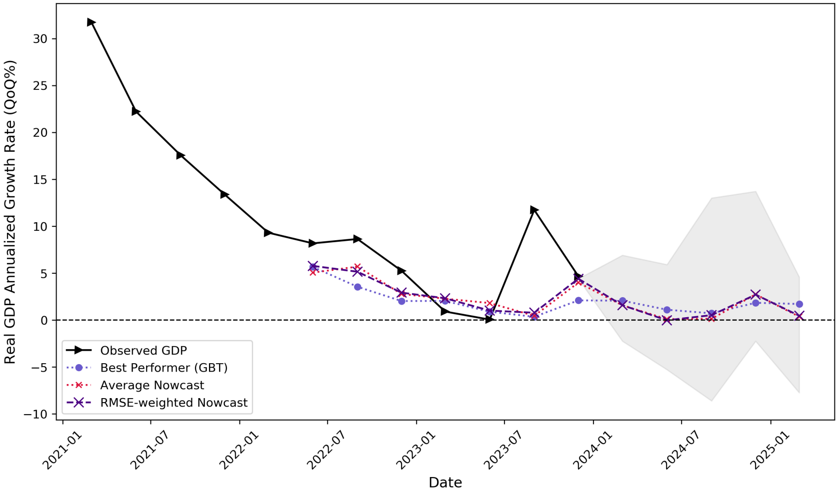
Inverted weight scheme keeping the vintage constant:

$$\omega_{iv} = \frac{1/\mathrm{RMSE}_{iv}^2}{\sum_j 1/\mathrm{RMSE}_{jv}^2}$$

Carry-on these weights to out-of-sample nowcast

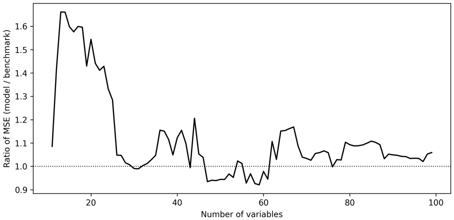
- Zero-back (V=0) and Two-ahead (V= +2) weighted nowcasts shown alongside observed QoQ % growth

Case Study: The Bahamas — Results — Out-of-sample nowcast range



Case Study: Jamaica — Variable selection (LARS)

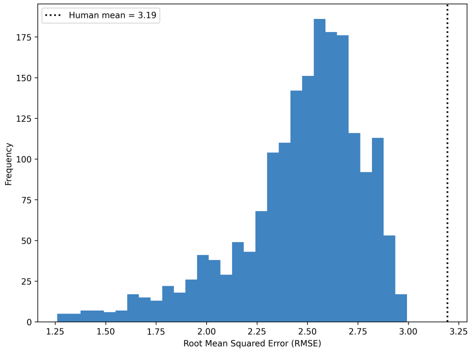
- LARS incrementally adds the most-correlated predictors
- Compare RMSE: 5-factor OLS with k vs. all-K variables; keep groupings beating the benchmark
- Country-specific groupings retained for Jamaica



LARS MSE-ratio profile vs benchmark (Jamaica)

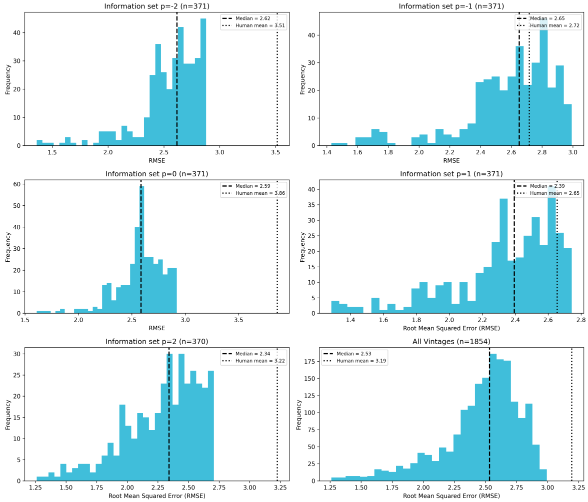
Case Study: Jamaica — Results — Distribution of estimates

- Dimensions: **5 vintages** $\times$ **7 estimators** $\times$ **34 specifications** $= N = 3{,}710$ estimates
- Preliminary RMSE distribution: **median 0.93 pp**, with a long tail (worst $\sim$4,000)
- We keep only estimates with RMSE below threshold ($N = 595$)



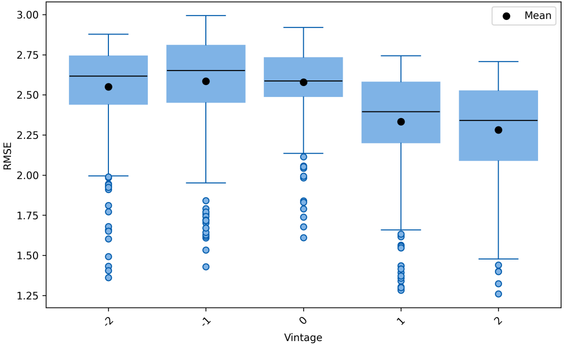
RMSE distribution (Jamaica) vs human-forecast mean

Case Study: Jamaica — Results — Distribution of estimates by vintage

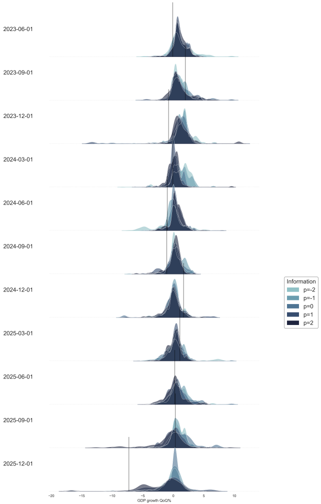


Case Study: Jamaica — Results — Estimate precision improves with information

- Precision tends to improve with each successive information set
- Average error drops from $\sim$**1.0 pp** to $\sim$**0.8 pp**

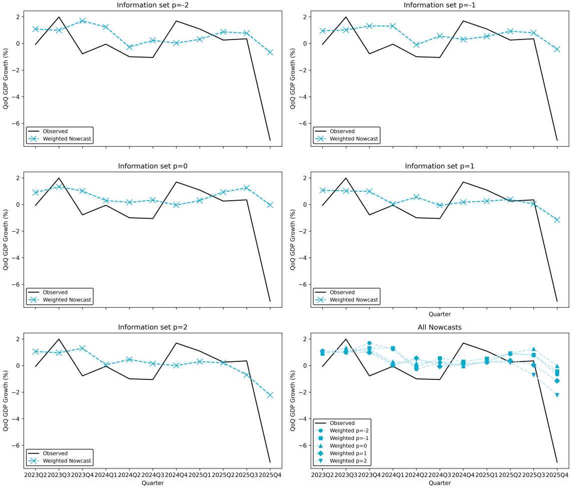


Case Study: Jamaica — Results — Distribution of Nowcasts



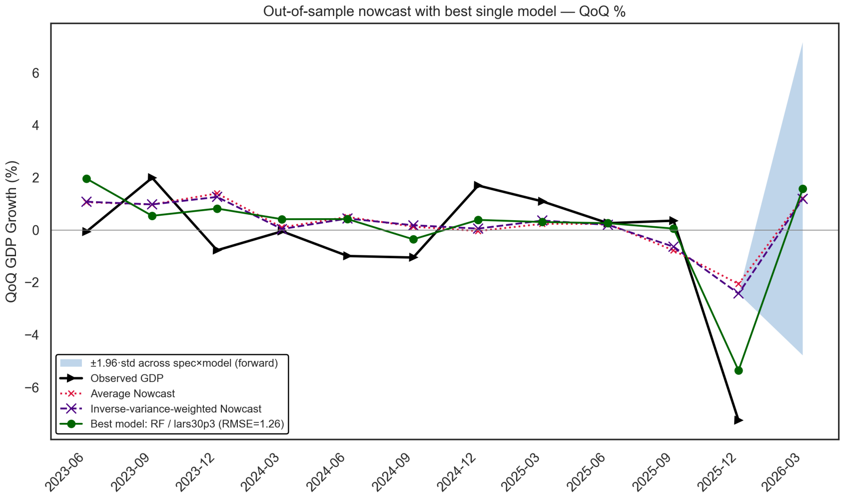
Observed Real QoQ % GDP growth vs nowcasts

Case Study: Jamaica — Results — Weighted nowcast — a first look



Estimates weighted by inverse RMSE

Case Study: Jamaica — Out-of-sample nowcast, Q1 2026



Barbados: Data integration

Comprehensive Dataset Structure (458 variables; 369 after filters)

Structured Data (367 series)

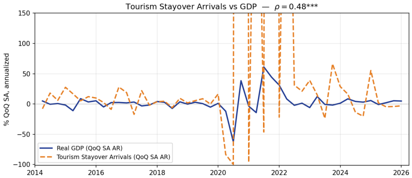
- **Tourism**: stayover, cruise, passenger arrivals
- **Trade**: imports & exports by category (US/CA/UK/EU)
- **Fiscal**: tax revenue (direct/indirect), grants, balance, debt
- **Financial**: loans, deposits, rates, soundness indicators
- **Monetary**: CPI, intl reserves, monetary base

Unstructured Data (91 series)

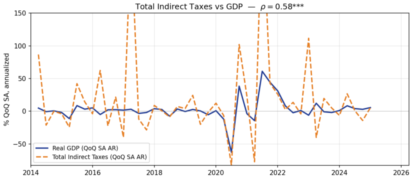
- **Google Trends** (32): tourism (hotels, beaches, all-inclusive), retail, activity
- **Satellite NTL** (59): hotels, ports, beachfront, near-nadir composites

Tourism receipts are ∼40% of GDP; structured indicators are anchored on tourism + trade flows, complemented by real-time digital signals.

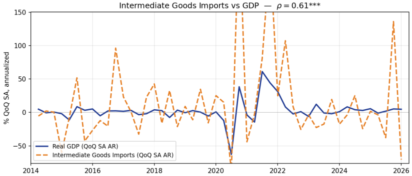
Barbados: Structured indicators vs GDP



(a) Tourism Stayover Arrivals



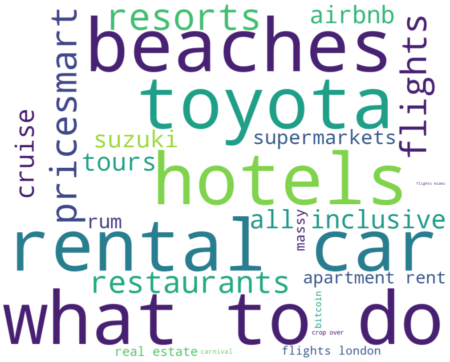
(b) Total Indirect Taxes



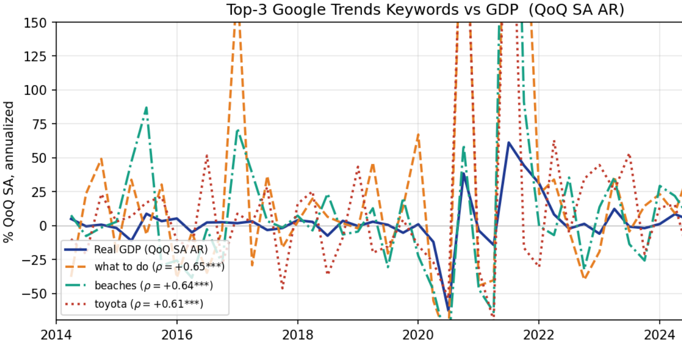
(c) Intermediate Goods Imports

(d) Consumer Goods Imports

Barbados: Unstructured data — Google Trends



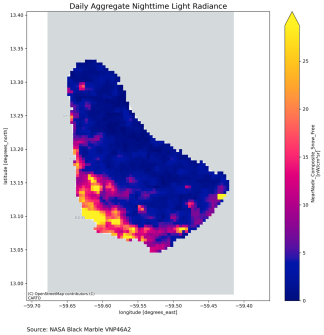
(a) Word cloud: keywords sized by $|\rho|$ with GDP



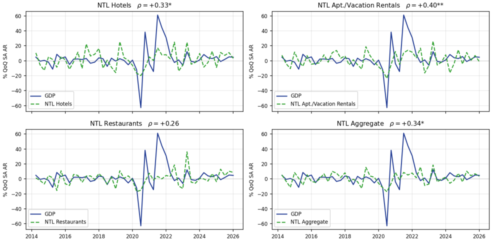
(b) Top-3 keywords vs GDP (QoQ SA AR)

26 variables so far (consumption, jobs, travel).

Barbados: Unstructured data — Nighttime Lights



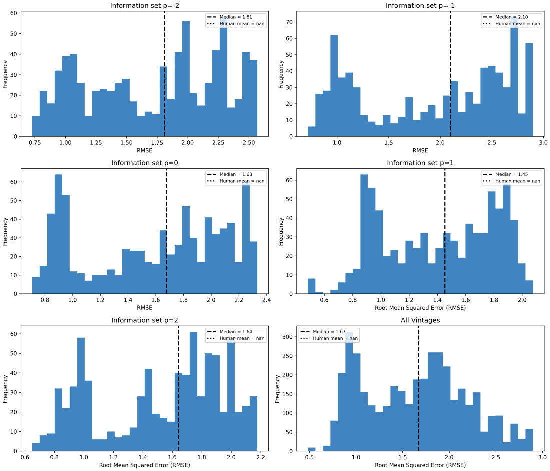
(a) Daily aggregate NTL radiance, Barbados



(b) NTL vs GDP growth, by location class

- Locations: hotels, apartment/vacation rentals, restaurants, ports (Bridgetown), airport (Grantley Adams)
- Daily VIIRS (NASA Black Marble VNP46A2)
- Co-moves with GDP via travel/tourism and consumption activity (Henderson et al., 2012; Fontaine et al., 2025)

Barbados: Preliminary Results — Nowcast distribution — RMSE

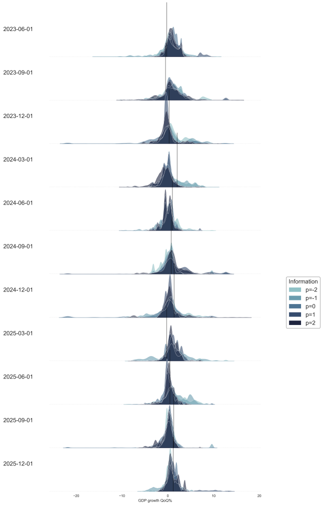


Barbados: Preliminary Results — ND improves performance

Model (QoQ %)	**RMSE**	**Improvement**
Naive (seasonal mean) benchmark	0.91	–
Top-quartile ensemble (V+0, current quarter)	0.86	**5%**
Top-quartile ensemble (V+1, post-close)	0.84	**8%**
Human-expert spec (OLSR, V+1)	0.85	**7%**
Best single model (DT, K=70, V+1)	0.48	**47%**

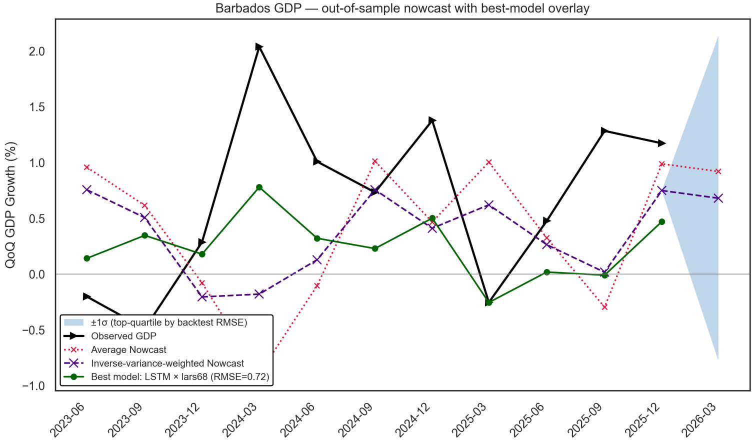
- Barbados target modeled in **QoQ %** with X-13ARIMA-SEATS seasonal adjustment
- Performance improves with information availability through V+0/V+1
- Top-quartile filtering identifies specs that match or beat the benchmark

- **Distributions of nowcasts** for best-performing models across vintages and test quarters
- Vertical lines indicate observed GDP growth
- Distributions contribute to addressing uncertainty



Barbados: Preliminary Results — ND ensemble improves accuracy over time

- Captures GDP **direction** across all vintages
- Under-amplifies magnitude
- Ensemble = median across 76 specs × 6 estimators × 3 seeds, per vintage
- p= −1 has the lowest median RMSE
- We are in the process of fine-tuning the model and your feedback is appreciated



Barbados: Concluding remarks

Core Innovation

Distribution of nowcasts rather than point estimates

Key Contributions

- First comprehensive nowcasting framework for a Caribbean economy
- Integration of unconventional data

Policy Impact

- 2–8 month lead on official GDP releases
- Real-time economic monitoring
- Scalable framework for data-constrained environments

Performance Results (Barbados)

- Naive seasonal-mean benchmark RMSE: **0.91** (QoQ %)
- Human-expert spec (OLSR, 8 hand-picked variables): RMSE **0.85** at V+1, **0.87** at V+0
- Top-quartile ensemble (LARS-driven, 15 specs): RMSE **0.84** at V+1, **0.86** at V+0
- Best single model: DT @ K=70, V+1, RMSE **0.48**
- Q1 2026 OOS: **+0.54%** $\pm$ 0.86 pp ($\pm 1\sigma$ top-quartile cross-spec dispersion)

Takeaways & Extensions

- The tool is useful to track economic activity in **real or near-real time**
- Strengthens the **policy dialogue** (fiscal–monetary, etc.)
- **Extensions**:
  - Region and/or sector nowcasting
  - Revenue and expenditure nowcasting
  - Sentiment analysis (textual signals)
- A tool to develop **capacities within the public sector**

Thank You

Q&A

Contact: jluissa@iadb.org

Inter-American Development Bank